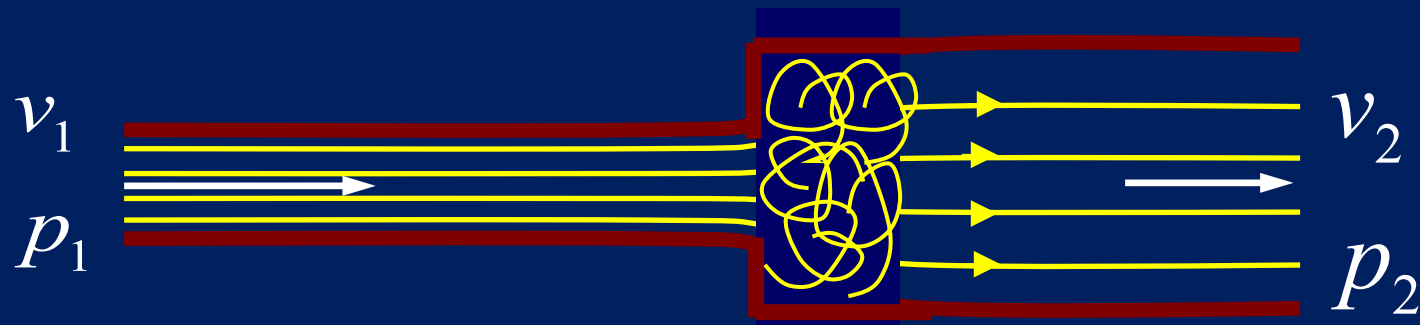
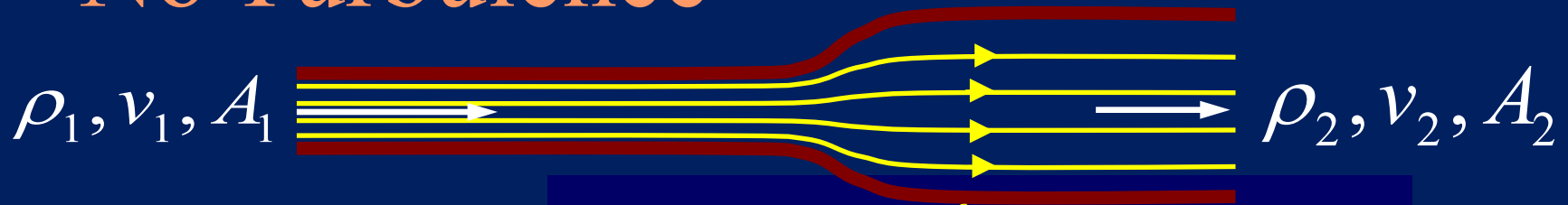


Quiz 11: Consider a stream of fluid of the density ρ with speed v_1 , passing abruptly from a cylindrical pipe of the cross-sectional area A_1 into a wider cylindrical pipe of the cross-sectional area A_2 . The jet will mix with the surrounding fluid and after the mixing, will flow on almost uniformly with an average speed v_2 . Without referring to the details of the mixing, Calculate the difference of the pressure at the two ends.

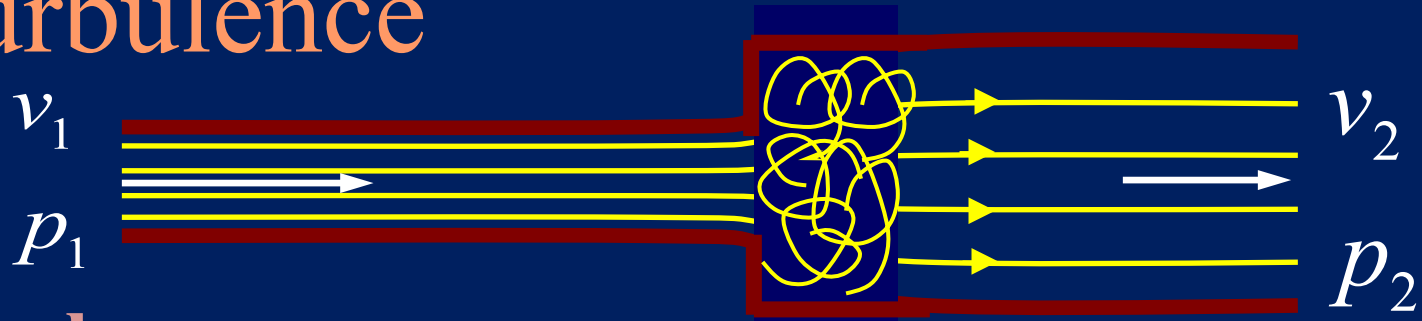


No Turbulence



$$p_2 - p_1 = \frac{1}{2} \rho (v_1^2 - v_2^2)$$

Turbulence



Impulse $J = \Delta P$

$$(p_2 - p_1) A_2 \Delta t = A_2 v_2 \Delta t \rho (v_1 - v_2)$$

$$p_2 - p_1 = \rho (v_1 v_2 - v_2^2)$$